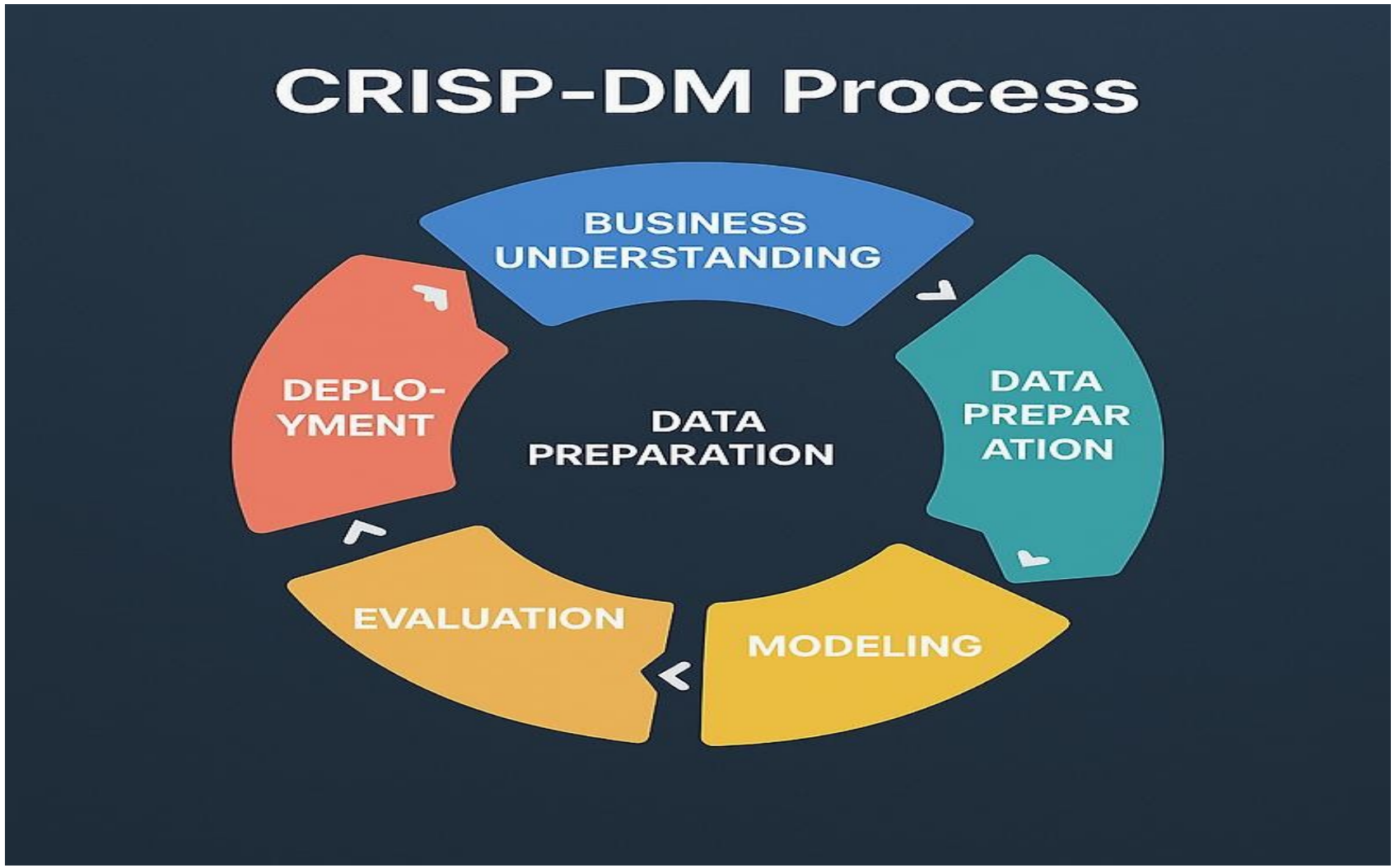
328-Introdution to Data Science

Class- 4

CRISP-DM

- Cross-Industry Standard Process for Data Mining.
- It is a structured framework for conducting data-mining and data-analysis projects.
- CRISP-DM has **six major phases**:
- **Business Understanding**
- **Data Understanding**
- **Data Preparation**
- **Modeling**
- **Evaluation**
- **Deployment**


CRISP-DM Process
BUSINESS UNDERSTANDING
DATA PREPAR ATION
MODELING
EVALUATION
DEPLO- YMENT
DATA PREPARATION

Data Science Lifecycle	**CRISP-DM**
Broad data-science framework	Structured data-mining framework
Includes data science activities	Six defined phases
Often includes communication and monitoring	Deployment is the final named phase
Can include advanced ML/AI workflows	Originally focused on data mining
Flexible	More standardized

Phase 1: Business Understanding

- The first phase focuses on understanding the **business or research problem**.
- **Problem:**
  A bank wants to identify customers who may leave the bank.
- **Objective:**
  Predict customer churn.
- **Success:**
  Identify high-risk customers early enough for the bank to take action.

Phase 2: Data Understanding

- Now we collect and examine the available data.
- **Example variables**
- Customer age
- Income
- Account balance
- Number of transactions
- Customer service calls
- Account duration
- Churn status
- We check:*Is the data sufficient and relevant to our problem?*

Phase 3: Data Preparation

- **Activities**

1. Data Cleaning
2. Data Transformation
3. Feature Selection
4. Feature Engineering

Example

Original:Date of Birth = 2000-05-15

We may create:Age = 26

Phase 4: Modeling

- Now we build models using prepared data.
- **Example**
- For customer churn:
- We may build several models and compare their performance using appropriate evaluation measures.
- **Example**
- A model has:
- **Accuracy = 95%**
- That sounds good, but suppose only 5% of customers actually churn.

Phase 6: Deployment

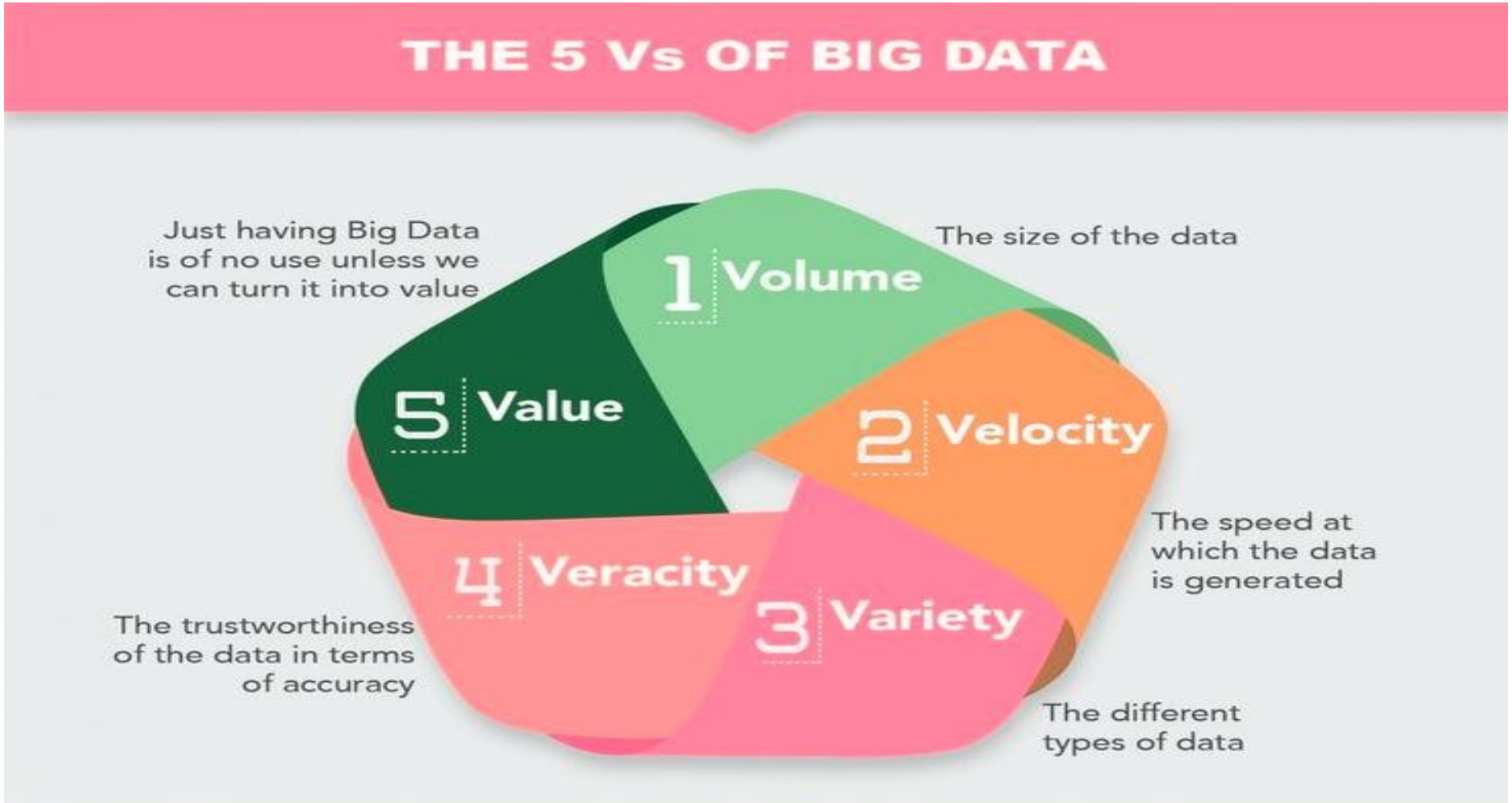
- **Phase 6: Deployment**
- The final model is put into practical use.
- Example
- A bank's system automatically calculates:Churn Risk = 0.87
- The customer-service team can then investigate that customer.

Big Data

- Big Data = Data that is too large, too fast, too diverse, or too complex for traditional methods to handle efficiently.

- **Example**
- A small shop may store customer purchases in an Excel file.
- A global e-commerce company may process **millions of transactions, searches, clicks, and product interactions**.

The 5 V's of Big Data



V	Example
Volume	Millions of trip and customer records
Velocity	GPS locations arrive continuously
Variety	GPS, text, payment, ratings, images
Veracity	Incorrect GPS or missing information must be handled
Value	Data helps optimize routes, pricing, and service

Technologies Used for Big Data

- **Storage and Processing**
- Hadoop
- Apache Spark
- Distributed file systems
- Cloud storage
- **Databases**
- MongoDB
- Cassandra
- HBase
- **Cloud platforms**
- AWS
- Microsoft Azure
- Google Cloud

VOLUME = How much?
VELOCITY = How fast?
VARIETY = What types?
VERACITY = Can we trust it?
VALUE = Is it useful?

Data Mining Functionalities

- Data mining functionalities describe what we want to discover or accomplish from the data.
- For example:
- Describe a group → **Characterization**
- Compare two groups → **Discrimination**
- Find relationships → **Association**
- Predict a class → **Classification**
- Predict a numerical value → **Prediction**
- Find natural groups → **Clustering**
- Find unusual records → **Outlier Analysis**

Major Data Mining Functionalities

- **Descriptive Mining**

Describes what is present in the data.

- Concept/class description
- Association analysis
- Correlation analysis
- Clustering
- Outlier analysis

- Predictive Mining
- Classification
- Prediction

Concept/Class Description

- It has two important forms:

1.Data Characterization

2.Data Discrimination

- **Data Characterization**

Data characterization provides a concise summary of the general characteristics or features of a target class of data.

- **Data Discrimination**

Data discrimination compares the characteristics of a target class with one or more contrasting classes.

Characterization	**Discrimination**
Describes one target group	Compares groups
Finds general characteristics	Finds differences
"What is this group like?"	"How is it different?"